Main

+ main(String[] args)

UserInterface

- Scanner scanner
- Controller controller
+ boolean running
- DateTimeFormatter formatter

+ startProgram()
+ createMovie()
+ deleteMovie()
+ searchMovies()
+ showAllMovies()
+ editMovie()
+ printMenu()
- readInt(prompt: String): int
- readInt(prompt: String, oldValue: Integer): int
- readMovieDate(prompt: String): LocalDate
- numberFormatExceptionMessage()
- getNextYear(): int
- readYesNo(prompt: String): boolean
- printMovieInfo(movie: Movie)
+ markAsSeen()
+ notSeenInTime()
+ loadMovies()
+ saveMoviesIfChanged()
+ sortBy()

«enumeration»
sortField

TITLE
DIRECTOR
YEAR
LENGTH
GENRE

Controller

- MovieCollection movieCollection
- FileHandler fileHandler
- boolean changed

+ addMovie(title: String, director: String, yearCreated: int,
  inColor: boolean, lengthInMinutes: int, genre: String)
+ getMovies(): ArrayList<Movie>
+ searchMovies(searchText: String): ArrayList<Movie>
+ editMovie(movie: Movie, title: String, director: String,
  yearCreated: int, inColor: boolean, lengthInMinutes: int,
  genre: String): boolean
+ deleteMovie(movie: Movie): boolean
+ markAsWatched(movie: Movie, date: LocalDate)
+ getMoviesNotWatchedForDays(days: int): ArrayList<Movie>
+ loadMovies(): int
+ saveMoviesIfChanged(): boolean
+ getSkippedLineCount(): int

MovieCollection

- ArrayList<Movie> movies

+ getMovies(): ArrayList<Movie>
+ addMovie(title: String, director: String, yearCreated: int,
  inColor: boolean, lengthInMinutes: int, genre: String)
+ addMovie(title: String, director: String, yearCreated: int,
  inColor: boolean, lengthInMinutes: int, genre: String,
  date: LocalDate)
+ addMovie(movie: Movie)
+ getNumberOfMovies(): int
+ deleteMovie(movie: Movie): boolean
+ editMovie(movie: Movie, title: String, director: String,
  yearCreated: int, inColor: boolean, lengthInMinutes: int,
  genre: String): boolean
+ editMovie(movie: Movie, title: String, director: String,
  yearCreated: int, inColor: boolean, lengthInMinutes: int,
  genre: String, date: LocalDate): boolean
+ searchMovies(searchText: String): ArrayList<Movie>
+ getMoviesNotWatchedSince(date: LocalDate): ArrayList<Movie>

FileHandler

- String fileName
- String HEADER
- int skippedLineCount

- toCsvLine(movie: Movie): String
+ saveMovies(movies: ArrayList<Movie>)
- parseMovie(line: String): Movie
+ loadMovies(): ArrayList<Movie>
+ getSkippedLineCount(): int

Movie

- String title
- String director
- int yearCreated
- boolean inColor
- int lengthInMinutes
- String genre
- int FIRST_MOVIE_YEAR
- LocalDate lastWatchedDate
- DateTimeFormatter formatter

+ getTitle(): String
+ getDirector(): String
+ getYearCreated(): int
+ isInColor(): boolean
+ getLengthInMinutes(): int
+ getGenre(): String
+ setTitle(title: String)
+ setDirector(director: String)
+ setYearCreated(yearCreated: int)
+ setInColor(inColor: boolean)
+ setLengthInMinutes(lengthInMinutes: int)
+ setGenre(genre: String)
+ getLastWatchedDate(): LocalDate
+ markAsWatched(date: LocalDate)
+ markAsWatched()
+ setLastWatchedDate(date: LocalDate)
+ hasBeenWatched(): boolean
+ getDaysSinceWatched(today: LocalDate): long
+ compareTo(other: Movie): int

«interface»
Comparable<Movie>

+ compareTo(other: Movie): int

«interface»
Comparator<Movie>

+ compare(movie1: Movie, movie2: Movie): int

TitleComparator

+ compare(movie1: Movie, movie2: Movie): int

DirectorComparator

+ compare(movie1: Movie, movie2: Movie): int

YearComparator

+ compare(movie1: Movie, movie2: Movie): int

LengthComparator

+ compare(movie1: Movie, movie2: Movie): int

GenreComparator

+ compare(movie1: Movie, movie2: Movie): int